

Dr. Himanshu Sen, PhD

Research Scientist | Molecular Biology | Microbiology

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Professional Summary

Molecular microbiologist with expertise in bacterial genetics, protein–DNA interactions, and regulatory protein biology. Experienced in molecular cloning, mutagenesis, protein purification, and biochemical assays to study transcriptional regulation and antimicrobial mechanisms. Proven record of peer-reviewed publications and independent research execution. Seeking research scientist, molecular biology roles in biotechnology and life sciences industries.

Key Expertise

- Bacterial and Yeast culture
 - Bacterial genetics & transformation
 - Protein–DNA interaction assays (EMSA)
 - Western blotting & biochemical assays
 - Molecular cloning & site-directed mutagenesis
 - Scientific data analysis (OriginPro, MS Office)
 - DNA sequencing & microbial identification (16S rDNA)
 - PCR, qRT-PCR
 - Antibiotic resistance profiling
 - Protein expression & purification
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Professional Experience

Senior Project Associate

CSIR–Institute of Microbial Technology (IMTECH), Chandigarh

Jul 2024 – Jul 2025

- Isolated environmental bacterial strains from outdoor polluted air samples in Delhi and established pure cultures for downstream analysis.
 - Performed microbial identification using **16S rDNA sequencing and phylogenetic analysis**.
 - Conducted **antibiotic resistance profiling** of pathogenic Staphylococcus isolates.
 - Evaluated **biofilm formation under environmental pollutant exposure** (black carbon particles).
 - Generated phenotypic data linking air pollution exposure to bacterial virulence traits.
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Doctoral Research

PhD – Biological Sciences

2019 – 2025

CSIR–Institute of Microbial Technology (IMTECH)

PhD Supervisor: Dr. Saumya Raychaudhuri

Thesis: *Study on the functional conservation and divergence among HapR and its homologues: lessons from Vibrio cholerae as a surrogate host*

- Investigated **structure–function relationships of quorum-sensing transcription regulators** in *Vibrionaceae*.
 - Designed **site-directed mutagenesis strategies** to probe DNA-binding residues in regulatory proteins.
 - Performed **protein purification and biochemical assays** to characterize regulatory activity.
 - Conducted **EMSA and protein–DNA interaction studies** to determine transcriptional regulation mechanisms.
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Education

PhD – Biological Sciences

2019 – 2025

CSIR–Institute of Microbial Technology (IMTECH), Chandigarh

MSc – Biotechnology

2016 – 2018

Indian Institute of Technology Roorkee

BSc (Hons.) – Microbiology

2013 – 2016

University of Delhi

Publications

1. Sen H, et al. Comprehensive findings of functional modulations linked ... Int J Biol Macromol. 2026.
2. Bhalerao SE, Sen H, et al. Administration of novobiocin and apomorphine ... Plos one. 2024.
3. Surin S, et al. Identification of critical amino acids in the DNA binding domain ... Plos one. 2024.
4. Sen H, et al. L-Ascorbic Acid Restricts Vibrio cholerae Survival in Various Growth Conditions. Microorganisms. 2024.
5. Sen H, et al. Diversity in the ligand binding pocket of HapR attributes to ... Int J Biol Macromol. 2023.
6. Ekka M, et al. Arginine 37 of Glycine Linker Dictates Regulatory Function of HapR. Front Microbiol. 2020.

7. Bhando T, et al. Antibacterial properties and in vivo efficacy of a novel nitrofuran, IITR06144, against MDR pathogens. J Antimicrob Chemother. 2020.
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Conferences

- Oral Presentation – International Conference on Harnessing the Power of Microbes, Chandigarh, India (2025)
 - Poster – MEEGID XVI, Dresden, Germany (2023)
 - Poster – ASCODD, Kolkata, West Bengal (2022)
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Fellowships & Awards

- SERB International Travel Support (2023)
 - CSIR Foreign Travel Grant (2023)
 - ASCODD Travel Grant (2022)
 - Best Oral Presentation Award (2025)
 - CSIR-UGC Junior Research Fellowship (Jul 2017, Dec 2017)
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Referees

Dr. Saumya Raychaudhuri, PhD
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